

DYNAMIC SYNTAX LABORATORY

Dynamic Syntax

Theoretical coverage and an implementation programme

A source-led map of construction families, semantic obligations, current workbench coverage and the next implementation stages.

18 categories; 91 tracked topics. Topics are research entries, not a count of implemented or independently verified analyses.

First implementation: named-head English nonrestrictive relatives through LINK, head copying, MERGE and supplemental assertion in Classical and Constructive DS.

Research and implementation audit: 6–7 October 2026.

Targeted literature review; version and evidence boundaries stated throughout.

Contents

1	Findings that change our development strategy	1
2	Research method and evidence boundaries	1
2.1	Search and selection	1
2.2	Evidence labels	2
2.3	Scope of the code audit	2
3	A mechanism-based taxonomy	2
3.1	C01. Core incremental growth and lexical programs	2
3.2	C02. Nominals, reference, quantification and scope	3
3.3	C03. Predication, valency and copular families	3
3.4	C04. Flexible order, case and discontinuity	3
3.5	C05. Relatives, resumption, extraction and islands	4
3.6	C06. Modification: adjectives, adverbs, PPs and comparison	4
3.7	C07. Coordination and shared material	4
3.8	C08. Anaphora, binding and contextual dependencies	4
3.9	C09. Clitic placement, clusters and person restrictions	5
3.10	C10. Auxiliaries, restructuring, tense, aspect and voice	5
3.11	C11. Clause linkage: causal, temporal, conditional and concessive	5
3.12	C12. Ellipsis and fragments	6
3.13	C13. Questions, split utterances, repair and grounding	6
3.14	C14. Information structure and peripheral constructions	6
3.15	C15. Agreement, plurality and typological tests	6
3.16	C16. Historical development and variation	7
3.17	C17. Learning, search and generation	7
3.18	C18. Semantic architectures and interfaces	7
4	Cross-backend implementation obligations	7
5	First implementation: supplemental relatives	8
6	Prioritized implementation programme	9
6.1	P0. Evidence and semantic regression infrastructure — started	9
6.2	P1. Relative family and reference — first English increment complete	9
6.3	P2. Contextual predicate reuse and ellipsis	9
6.4	P3. Nominal coordination, possessives and common determiners	9
6.5	P4. Events, auxiliaries, voice and selected complements	10
6.6	P5. Questions, answers and agreement	10
6.7	P6. Clause semantics and discourse attachment	10

6.8 P7. Wider typology and empirical scale 10

7 A meaningful role for the LLM and Jev 10

8 Evaluation: what broad coverage must mean 11

9 Bibliography and review ledger 11

9.1 Sources to acquire or review directly next 13

10 Relation to the project foundation 13

Research and source-code audit, 6–7 October 2026. This is a first substantial evidence map and a working implementation programme. It distinguishes a published analysis, a proposal, an implemented fragment, and a demonstrated meaning. These are different achievements.

The companion `ds-theoretical-coverage.json` is the machine-readable catalogue. The bibliography below identifies the versions actually inspected. Source keys such as [CKM05] refer to those entries. A source’s discussion of a phenomenon does not make that phenomenon implemented in this workbench.

1 Findings that change our development strategy

DS has a substantial theoretical literature extending far beyond incremental subject–verb–object parsing. It offers analyses of discontinuity, restrictive and nonrestrictive relatives, resumption, scrambling, agreement, clitic placement and person restrictions, copular constructions, ellipsis, and interactive dialogue. There is also computational work on grammar induction, generation, semantic updating, and feedback.

The central organizing principle is **reusable mechanisms with language-specific lexical conditions**. A catalogue of isolated words such as *because*, *when*, or *who* is therefore the wrong development unit. A useful unit is a construction family with specified tree operations, semantic composition, context assumptions, locality conditions, and failure controls.

Four findings are particularly consequential for this project:

1. **LINK is not a single semantic operation.** Restrictive relatives, nonrestrictive relatives, coordination, topic structures and adjuncts can all use LINK, but their evaluation differs. Reusing an edge type without specifying how meanings combine does not implement the analysis.
2. **DS is not inherently TTR.** The 2021 overview explicitly distinguishes tree-growth machinery from the representation of semantic content [HG21, §2.1, p.265]. Classical terms, TTR records, distributional semantics and the later Constructive proposal represent different choices. Sharing a parser does not establish semantic or grammatical parity.
3. **Theoretical breadth is not an off-the-shelf broad-coverage grammar.** The 2011 DyLan paper reports an implemented range of constructions, but also a small lexicon and limited coverage [PEH11, §§3.5]. That historical implementation claim cannot simply be inherited by a Python port or by a different semantic backend.
4. **A valid derivation is insufficient evidence of the intended meaning.** Type correctness, referent identity, argument roles, quantifier scope, modifier attachment, contextual licensing and lexical sense all need separate checks. Our LLM-assisted system currently addresses only some of these obligations.

The first implementation from this map is now a named-head nonrestrictive relative family in native English Classical and Constructive DS. It follows the explicit LINK/copy/MERGE/conjunction analysis in [CKM05, §3.1] and [HG21, §2.5.1]. English restrictive relatives already existed; conflating their restrictive semantics with the new supplemental assertion would be an error. Greek relatives and a TTR parity audit are separate next work items.

2 Research method and evidence boundaries

2.1 Search and selection

The review began with the core DS book, the four-dialect Greek-clitic thesis, and their references, then added the 2021 framework overview, published DS-TTR implementation papers from ACL Anthology, the ellipsis chapter, interaction work, modification proposals, and the 2025 Constructive article. Crossref was used for publication metadata and discovery. Author bibliographies and references supplied further

leads. The local archive contained both complete documents and empty cloud placeholders; only readable documents were used as evidence.

Direct sources include a complete 386-page prepublication book, a 396-page thesis, journal articles, chapters, and computational papers. Review was targeted: contents and relevant chapters/sections were inspected, not every page of every document. Full source texts remain outside the repository; the catalogue records retrieval locations and review scope. The bibliography includes publicly resolvable links where available.

This is a structured, citation-led review, **not an exhaustive systematic review of every DS publication**. Search engines and the project website were not consistently accessible; primary publisher/ACL downloads and the local archive provided the usable evidence. A 2026 handbook chapter was identified through metadata but not reviewed. Its existence is recorded as a follow-up, not treated as proof of additional construction coverage.

2.2 Evidence labels

Label	What it establishes
Worked analysis	Inspected rules, derivation, or explicit semantic treatment for the relevant mechanism. It can still cover a small fragment.
Implemented study	A source reports executable parsing, generation or learning. Its original conditions and evaluation remain attached to the claim.
Proposal or sketch	A mechanism is motivated or outlined; details or a full derivation remain open.
Secondary pointer	A reviewed source identifies another analysis; that underlying publication still needs direct inspection.
Metadata only	Bibliographic existence is verified; substantive claims are not drawn from it.

Implementation labels are independent: **tested fragment**, **partial infrastructure**, **not implemented in the native workbench**, and **not yet audited**. In particular, “not yet audited” is the right label for many inherited TTR capabilities. It must not silently mean either full support or absence of support.

2.3 Scope of the code audit

The implementation map uses maintained grammar files, atomic effects, design notes and tests. Important locations include `relative_effects.py`, `semantic_effects.py`, `causal_effects.py`, `native_dialogue.py`, the native grammar directories, the 2015–english–ttr grammar, and the semantic modules. Older coverage plans are historical evidence, not current feature inventories. The catalogue’s file links provide starting points for reproducing the audit.

3 A mechanism-based taxonomy

The following eighteen categories separate construction families from their semantic representations. Languages are those discussed in the cited analyses; they are not a list of languages the app currently parses.

3.1 C01. Core incremental growth and lexical programs

Theory. Types and formulae decorate binary semantic trees. Requirements state what remains to be built. Lexical programs and computational operations update the pointed tree. Introduction, prediction, anticipation, completion, thinning and elimination provide basic growth [CKM05, ch.2; HG21, §§2.1–2.3].

Current implementation. The package supplies tree operations, action parsing, requirements, a context DAG and backtracking. Native semantic effects add backend-specific type checking and composition. Recorded operations drive the browser visualization.

Next obligation. Treat action traces as testable derivation evidence. Check the intermediate type and formula state, not only the final result. Failure under a bounded incomplete grammar is not an independent judgment that a natural-language sentence is ungrammatical.

3.2 C02. Nominals, reference, quantification and scope

Theory. [CKM05, §§3.2–3.3] develops structured nominal terms, restrictive content, epsilon/tau-style term operators and scope constraints. [C25, §§3–4.2] replaces this semantic machinery with many-sorted dependent types, Σ/Π and coercions. These are materially different analyses, not alternate print styles for one formula.

Current implementation. Names, selected determiners, singular count nouns, contextual definites, intersective refinements and limited quantifier scope work in native English. Distributive *all* with plural forms denotes individual-level quantification. Greek has a smaller case-sensitive DP inventory. General possessives, mass/count interpretation, numerical quantification, bare plurals and collective/cumulative readings remain gaps.

Next obligation. Separate reference identification from existence assertions. A contextual definite must not become a fresh arbitrary constant without an exposed assumption. Build tests that distinguish one shared existential from a separate existential per universal.

3.3 C03. Predication, valency and copular families

Theory. DS lexical programs project predicate–argument structure. [CKM05, ch.8] treats predicative, existential *there be*, equative and specificational copulas through underspecification and contextual update. These are several related constructions, not one semantically empty *be*.

Current implementation. Native active intransitive/transitive and selected ditransitive, dative, clausal and PP-complement frames work. Nominal and adjectival copulas and selected locations are present. Corpus/WordNet occurrences propose frames; they do not establish exhaustive lexical valency. General existential *there*, equative and specificational analyses are absent from the native coverage claim.

Next obligation. Implement *there be* as a sourced referential and predicational analysis with explicit contextual conditions. Merely ignoring *there* and accepting the remaining noun phrase would not reproduce ch.8.

3.4 C04. Flexible order, case and discontinuity

Theory. Unfixed positions, local and general adjunction, case filters and MERGE capture delayed placement without moving a previously fixed constituent. [CKM05, §§2.4–2.5, ch.6] develops this for dislocation and Japanese scrambling. [HG21, §2.4] explains why two nodes with the same unfixed address cannot remain distinct.

Current implementation. Unfixed nodes and address constraints exist; reviewed Greek case-marked names support several orders. A finite set of working orders is not a complete scrambling grammar.

Next obligation. Test incremental availability and incompatible case combinations. Add multiargument and embedded cases only with explicit locality conditions. Japanese/Korean analyses inform mechanisms but are not evidence that identical Greek or English lexical programs are correct.

3.5 C05. Relatives, resumption, extraction and islands

Theory. [CKM05, chs.3–4] gives nonrestrictive and restrictive relatives, crossovers, and cross-language resumption patterns. A LINK can originate at an identified individual or inside a nominal restriction. The copied term and evaluation point differ. The dominance relation used for a gap does not automatically cross LINK boundaries [CKM05, §3.1.2, p.80].

Current implementation. Native English has restrictive subject/object gaps with LINK, head copying and MERGE. This increment adds named-head nonrestrictives. The existing restriction to local subject/direct-object gaps remains. Greek *nov*, clitic resumption in relatives, PP gaps and unbounded extraction are not covered by these entries.

Next obligation. Greek *nov* must interact with case, pro-drop and clitics; it cannot be an English *who* alias. English nonlocal relatives require explicit intervening-clause and island tests. Do not turn off locality checks to make a difficult example pass.

3.6 C06. Modification: adjectives, adverbs, PPs and comparison

Theory. There is no single settled DS implementation for all modifiers. [C17] explicitly describes the earlier literature as sparse and contrasts in-tree adjectival refinement with LINK treatments. It proposes different attachment levels for sentence, VP and temporal adverbs. [C25, §§4.4–4.5] works through dependent adjective refinements and adverbial LINK options. [PEH11, §2.1] demonstrates event-related record extension in DS-TTR.

Current implementation. Native intersective adjectives, selected adverbs, PP argument frames, PP modifiers and nominal comparisons exist. Selected arguments satisfy a verb's explicit requirements; modifiers build additional LINK structure. Present temporal modifiers retain an opaque contribution; they do not supply a full event/time calculus.

Next obligation. Establish event and situation semantics before claiming general temporal, manner, agent-oriented or speech-act modification. Preserve attachment alternatives. Subjective/intensional adjectives, degree semantics and clausal comparison need separate treatment from intersective adjectives and nominal similes.

3.7 C07. Coordination and shared material

Theory. [CKM05, §§5.4,7.3–7.4] combines LINK, contextual sharing and late adjunction for coordination, right-node raising and agreement effects. Right-node raising involves additional assumptions about intonationally licensed update; it is not demonstrated merely by a conjunction lexical item.

Current implementation. Selected clause and VP coordination work in the native fragments. General NP coordination, gapping, right-node raising, shared complements and their number interpretation are not implemented as a full family.

Next obligation. Separate proposition conjunction, predicate conjunction, group-forming nominal coordination and shared-argument composition. Use positive/negative pairs that expose argument duplication, omission and accidental sharing.

3.8 C08. Anaphora, binding and contextual dependencies

Theory. Metavariables receive content through constrained substitution [CKM05, §§2.5,3.1.3–3.1.4; HG21, §2.4.3]. Reflexives impose local conditions; personal pronouns can depend on wider context. Context can include partial trees and action histories, not just a list of previously mentioned names.

Current implementation. Selected speaker/hearer references, named antecedents and local English reflexives are available. The app exposes assumed antecedents. This does not establish discourse salience, donkey anaphora, plural antecedents, arbitrary logophoricity or binding across every construction.

Next obligation. Store multiple antecedent candidates with provenance, number/person/case constraints and scope accessibility. A model may rank compatible candidates; it must not erase binding constraints or turn a guessed antecedent into a verified fact.

3.9 C09. Clitic placement, clusters and person restrictions

Theory. [C10, chs.3–6] analyzes Standard Modern, Cypriot, Grico/Grecia Salentina and Pontic Greek. [CK11] develops person restrictions through the timing of fixing underspecified structure. Case can trigger immediate structural enrichment before another unfixed node is introduced [CK11, pp.144–145].

Current implementation. The workbench contains reviewed placement fragments for four varieties, with more developed SMG/Grico clusters, ditransitives and strong PCC controls. Those supported lexical programs do not cover all claims or variants in the thesis.

Next obligation. Extend by construction environment and source example, including contrasting person combinations. Dialect labels must identify distinct grammars rather than spelling substitutions. A shared finite-verb template does not establish a shared clitic system.

3.10 C10. Auxiliaries, restructuring, tense, aspect and voice

Theory. The core book explicitly sets passive/progressive aside at the start of ch.8. Later work must therefore be consulted. [C10, §3.4.3.2, pp.143–146] develops clitic climbing by treating auxiliaries/restructuring predicates as contributions to a complex situation argument rather than ordinary independent biclausal predicates. It refers to the work later published as Cann’s 2011 English auxiliary chapter. [KC16] also points to that chapter for passive-related structural conditions.

Current implementation. Native do-support, selected modal operators, Greek finite forms and several particles do not amount to tense/aspect or voice semantics. General passive, progressive, perfect, raising/control and restructuring are not in the native broad-coverage claim.

Next obligation. Obtain and inspect Cann (2011) and the relevant restructuring derivations in full. Build an event/situation representation, then implement auxiliary sequences and role mappings. Do not implement passive simply by swapping two strings in an active predicate.

3.11 C11. Clause linkage: causal, temporal, conditional and concessive

Theory. The literature includes a dedicated conditional thesis (Gregoromichelaki 2006, identified in [HG21] and [KE19]) and treatments of Greek subordination and clitic triggers [C10, §§4.6–4.7]. These establish a research route, not equivalence between every connective and conjunction.

Current implementation. The app now has a reusable finite-clause family with distinct causal, temporal, conditional, concessive and contrastive relations, preposed/postposed orders and local content closure. Relations are opaque, typed predicates over contents. They do not compute temporal order, counterfactuality, causation, concessive expectations or conditional entailments.

Next obligation. Review the conditional analysis directly. Connect construction choices to a developed semantics and context model. Until then, LLM selection of *since* as temporal or causal remains a lexical/contextual hypothesis, even when the DS derivation completes.

3.12 C12. Ellipsis and fragments

Theory. [CKM05, §9.2; KE19, §§1.2–1.3; PEH11, §3.2] distinguish formula reuse, action rerun and continuation of an existing partial tree. These support strict/sloppy VP ellipsis, fragment answers, corrections and other contextual dependencies. [C25, §4.3] proposes a typed history of DS actions and an explicit rerun example.

Current implementation. Shared continuation and local repair exist, but they are not a general implementation of VP ellipsis, sluicing, nominal ellipsis or strict/sloppy alternatives. The native pipeline does not yet retrieve and validate a typed reusable predicate history for arbitrary *does too* examples.

Next obligation. Begin with named-subject predicate reuse, then distinguish strict formula reuse from sloppy action rerun. Freshen witnesses and rebind indexicals correctly. An ellipsis resolver must fail when no licensed antecedent is available; it must not invent missing content with an LLM.

3.13 C13. Questions, split utterances, repair and grounding

Theory. [KC16; KE19; E15] analyze dialogue through incremental structure, action histories and participant-dependent contexts. [E15, §§2.3–3.2] adds self/other coordination pointers to model feedback and clarification. These are distinct from the semantic tree’s construction pointer. The model’s *no* can signal branch abandonment, not necessarily propositional negation.

Current implementation. Native English supports a bounded shared polar question/reflexive/answer fragment, speaker changes, continuation and local repair. The screen’s shared tree is not a complete model of two participants’ grounding states or knowledge. Wh-question interpretations, clarification subtypes and nonlinguistic grounding are mostly missing.

Next obligation. Make context histories and participant commitments explicit. Add short-answer and clarification families with answer-type and binding checks. Preserve repaired material and distinguish a new proposition from continued growth of the current tree.

3.14 C14. Information structure and peripheral constructions

Theory. [CKM05, §§4.2,5.1–5.3] analyzes left topics, focus-related alternatives, right dislocation, extraposition and late adjunction. [HG21] provides secondary pointers to Japanese/Rangi clefts and Japanese multiple focus analyses.

Current implementation. Initial named Greek topics have a small executable fragment. General hanging topics, afterthoughts, clefts and prosodic licensing are not broadly implemented. A structural strategy label in the UI is not an independent topic/focus or discourse-felicity judgment.

Next obligation. Pair form with discourse assumptions and information structure. Keep structural alternatives available when the same word order supports different analyses. Read the cleft papers before generalizing from the topic fragment.

3.15 C15. Agreement, plurality and typological tests

Theory. [CKM05, ch.7] derives order-sensitive full and partial agreement for Swahili conjoined nominals; singular/plural and noun-class information interact with when structure becomes fixed. [HG21, §§2.4,3] points to Rangi auxiliary order, Japanese case, Korean incrementality, Kazakh differential object marking and Mandarin minimizers.

Current implementation. These languages are not active workbench grammars. Selected Greek person/number/case restrictions and English distributive plurals are much smaller claims than full agreement and plurality.

Next obligation. Use these analyses as stress tests of general mechanisms. Do not introduce universal rules from English or Greek alone. Distinguish lexical morphological annotation, agreement licensing and semantic plurality. The secondary pointers need direct-source review before implementation.

3.16 C16. Historical development and variation

Theory. [C10] connects changing clitic placement to routinization and generalized parsing triggers. [HG21] identifies related work on Greek and Spanish change. These are explanatory historical proposals with empirical corpus conditions, not executable grammars for every historical stage.

Current implementation. Historical panels explain a sourced account; the four modern fragments are executable. The panels should not be read as demonstrations that Koine or medieval sentences can be parsed generally.

Next obligation. Encode a stage-specific lexicon and distributional evidence before offering historical parsing. Distinguish alternative historical hypotheses and register/period constraints.

3.17 C17. Learning, search and generation

Theory. [EPH13] learns lexical action hypotheses from utterances paired with semantic trees, using DAG intersections and probability estimates. [ECH13] extends the research to child-directed dialogue data. [PEH11; E15] describe parsing/generation over shared incremental representations.

Current implementation. The app searches a reviewed action inventory, optionally expands lexical entries with bounded model proposals, and uses Jev for lexical preferences. This is not the grammar induction algorithm from the literature. Source lookup is not training, and a replay accepting an alternative is not a learned grammar update.

Next obligation. Establish an adjudicated semantic dataset before any learning claim. Compare dictionary-only, model-assisted and Jev-ranked runs with matched inventories and budgets. Report accuracy separately from completion, abstention, latency and cost.

3.18 C18. Semantic architectures and interfaces

Theory. [HG21, §2.1] makes representational flexibility explicit. [PEH11; E15] show TTR records, manifest fields, dependencies, subtyping and event-related enrichment. [C25] develops nominal types, coercions, dependent quantification and action contexts. [KV19] is a short distributional-semantics proposal; it is not a specification for our native backends.

Current implementation. Classical and Constructive modes share execution but compile separate native meanings. The TTR path uses its own record algebra and inherited grammars. There is no general translation from completed native formulae into faithful TTR analyses, or vice versa.

Next obligation. Define a semantic contract per construction and backend: roles, reference identity, dependency scope, available inference and retained partial information. “Same sentence accepted” is weaker than equivalence.

4 Cross-backend implementation obligations

Area	Classical and Constructive obligations	TTR obligation
Nonrestrictive relative	Share the identified head; keep supplemental assertion separate from nominal restriction; close local witnesses.	Reuse the head's referent/path; preserve dependencies when merging records; avoid freshening the shared individual away.
Restrictive relative	Classical CN predicate conjunction versus Constructive dependent nominal refinement.	Extend the appropriate restrictor record before reference/quantification; do not flatten it into matrix assertions.
Ellipsis	Typed predicate reuse or validated action rerun; distinguish binding and witness freshness.	Preserve record dependencies and intended shared fields while freshening only new referents/events.
PP/modifier	Identify selected slot versus adjunct; preserve attachment and eventual event/situation scope.	Add role/event constraints to the right record or LINK structure; record extension alone must not choose an attachment silently.
Clause relation	Distinct locally closed contents; event, modal and discourse semantics where claimed.	Keep clauses and event identities distinct; do not flatten dependent clauses into one undifferentiated record.
Dialogue	Participant-sensitive reference and accessible action histories; explicit repair boundary.	Record/context updates plus participant coordination state; distinguish tree pointer from grounding pointers.

TTR is not an inferior or intrinsically nonconstructive interpretation. Likewise the workbench's label "Constructive" names a specific implemented MLTT-inspired proposal; it is not a claim that every other type-theoretic approach lacks constructive content.

5 First implementation: supplemental relatives

The initial implementation follows [CKM05, §3.1.1–3.1.2, especially (3.6), (3.9), (3.13)]. The source uses "John, who Sue likes, smokes." Our existing vocabulary gives an analogous executable example:

```
John, who Mary knows, walks.
main:      walk(john)
supplement: know(mary, john)
result:    walk(john)  $\wedge$  know(mary, john)
```

The opening comma starts a LINKed proposition from the identified head. The relative pronoun copies the head to an unfixed node; MERGE fixes it at the subject or direct-object position. A closing delimiter requires the relative to be complete before returning to the matrix. Final LINK evaluation conjoins the propositions. The original relative subtree remains inspectable.

In Constructive mode an internal indefinite stays local:

```
John, who knows a woman, walks.
walk(john)  $\wedge$   $\Sigma$  x:woman. know(john, x)
```

It must not become Σ x:woman. (walk(john) $\wedge$ know(john,x)) merely because the engine carries witnesses while composing. The implementation closes each proposition before conjunction. Classical mode instead retains its local epsilon term. The Constructive adaptation is ours, tested using the existing native term system; the Classical source is not misrepresented as an MLTT specification.

The tests cover both gap positions and both matrix argument positions; negation and modifiers in the matrix or relative; local indefinites; shared identity; recorded LINK/unfixed/MERGE/return operations;

malformed and unsupported cases; and alternative complete analyses. Constructive exports are compiled with Coq when installed. This checks exported typing, not empirical truth or grammatical acceptability.

The implementation also fixes propagation of an adverb's updated meaning: recomputation stops at a relative's LINK root instead of walking through the inverse LINK and erasing the nominal host. Sealing after supplemental evaluation prevents a later return to the original matrix VP from discarding the extra assertion.

Scope is explicit: main-clause identified constant heads, local subject/direct object gaps, *who/which*, written delimiters, supported negation/adverbs and internal indefinites. Quantified hosts, general definite descriptions, Greek *πou*, extraction across embeddings, animacy selection and supplement projection through content-taking clauses remain outside this increment. The latter combinations are blocked rather than assigned an unchecked projection analysis. See docs/design/nonrestrictive-relatives.md.

6 Prioritized implementation programme

6.1 P0. Evidence and semantic regression infrastructure — started

Maintain this catalogue as the single theory-to-code index. Every feature gets a source key, precise passage, theoretical status, target language, backend obligations, context assumptions, positive meanings and negative controls. Store authored examples separately from published judgments and held-out evaluation data. Existing small coverage numbers remain historical baselines; this review does not update them by assertion.

Exit criterion: a proposed coverage claim can be traced from a source to an executable derivation and its expected meaning, without reading a chat log.

6.2 P1. Relative family and reference — first English increment complete

Follow with a sourced Standard Greek *πou* fragment, including head case, clause-internal role and pro-drop/clitic interactions. Extend English nonrestrictives only after specifying definite/quantified hosts and projection. In parallel as a workstream, audit existing TTR relative programs against the same identity and role cases; this is a plan for subsequent work, not a claim that this audit has already been performed.

Exit criterion: meaningful positive and locality/binding controls in both native systems for each advertised language; separate TTR results reported.

6.3 P2. Contextual predicate reuse and ellipsis

Implement an accessible typed predicate-history interface. Begin with *John knows Mary. Bill does too.* Then add reflexive/possessive contrasts that distinguish strict from sloppy readings. Use [KE19], [PEH11] and [C25, §4.3], not a free-text model completion.

Dependencies: context accessibility, variable freshening, scope-aware reference, repeatable action replay.

Exit criterion: correct changed subject, preserved object, no witness leakage, and refusal without an appropriate antecedent.

6.4 P3. Nominal coordination, possessives and common determiners

Develop a number/reference representation that can support coordinated NPs without confusing them with proposition conjunction. Review fuller nominal analyses for possessives, numerals, mass/count interpretation and plural readings. The Swahili agreement chapter supplies valuable structural tests, but does not by itself specify all English/Greek nominal semantics.

Exit criterion: distinguish distributive and collective interpretations where supported, and do not accept the latter under an individual-only encoding. Keep unsupported reading classes visible.

6.5 P4. Events, auxiliaries, voice and selected complements

Directly review Cann (2011), Marten (2002), and the restructuring/clitic climbing sections. Implement a minimal event/situation layer with explicit reference-time assumptions. Add perfect/progressive/passive families and case/role mappings on that basis. Distinguish control, raising and ordinary finite complementation rather than assigning one generic clausal frame.

Exit criterion: tense/aspect contributions survive composition; passive agent/theme roles are checked; incompatible auxiliary combinations fail; native and TTR semantics each satisfy their own contracts.

6.6 P5. Questions, answers and agreement

Broaden wh-dependencies, short answers and clarification using context histories. Extend Greek person/number/case coverage and English agreement where it conditions acceptance. Use minimal contrasts and independently sourced restrictions. Dialogues must be evaluated as sequences with speaker and context state, not only as concatenated sentences.

6.7 P6. Clause semantics and discourse attachment

Replace unsupported semantic claims with developed analyses of temporal, conditional and causal content. Inspect the conditional thesis directly. Add argument-versus-adjunct and attachment alternatives to evaluation. General causal reasoning or counterfactual inference should not be advertised because the syntax accepts an opaque condition(main, antecedent) term.

6.8 P7. Wider typology and empirical scale

Expand dialect and historical fragments from source judgments; directly review the Rangi, Japanese, Korean, Kazakh and Mandarin pointers before coding them. Use the resulting diversity to test the general machinery. Scale lexicon coverage and search only after constructing reliable semantic tests.

Priority is based on reusable coverage and current prerequisites, not on a claim that every missing analysis is equally mature. P2–P6 can be developed in separate feature branches as resources permit, but each family retains its own entry and exit criteria.

7 A meaningful role for the LLM and Jev

The larger models are useful for proposing lexical analyses, identifying candidate construction families, retrieving relevant evidence, suggesting counterexamples and diagnosing a failed derivation. They do not supply a verified DS analysis simply by producing a plausible description.

The proposed operational path is:

1. Inspect tokens, current tree, accessible context and the reviewed construction inventory for the selected language/backend.
2. Propose a typed lexical entry, an existing construction/reading choice, or an explicit unsupported-construction diagnosis with evidence.
3. Validate that proposal against the selected compiler and grammar contract.
4. Execute DS and inspect requirements, identity, role and scope invariants.
5. Retain alternative analyses, uncertainty, sources, model calls and failed attempts in the result.

For new grammar development, an LLM can draft action programs and tests in the repository, but adding a reviewed family is a software/research change, not something silently approved during an end user's parse. Runtime model proposals remain bounded by the implemented inventory.

Jev's current useful role is to rank compatible lexical senses as context arrives. Future uses include ranking compatible attachment/antecedent candidates and deciding when to abstain, after calibration. These would be new features. The current implementation must not be described as already providing them. Typed probabilities are preferences, not proofs of truth, grammaticality, or correctness of the chosen DS analysis. The new nonrestrictive relative family works without the LLM because the missing resource was a construction program. With that program present, existing lexical assistance can supply eligible content words. This is how the formal grammar and models can contribute different, measurable things.

8 Evaluation: what broad coverage must mean

Use three distinct datasets:

- **Source-linked construction suites:** positive meanings and sourced contrasts, with dialect/context conditions. Authored regressions are marked as such. These establish targeted implementation behavior.
- **Development text:** natural sentences/passages used while engineering the grammar. Useful for error analysis; not an unbiased coverage estimate.
- **Frozen held-out text:** independently sampled English and Greek, with declared sampling unit, source/register, license and split. Do not add failed test sentences to the development suite and continue calling the same set held out.

Report all of the following with denominators: token lexical coverage, sentence derivation completion, whole-paragraph completion, semantic correctness for adjudicated items, ambiguity/abstention, cap/timeouts, latency, provider failures and model cost. Unknown words and unsupported constructions remain failures in overall coverage; report their categories separately. A later complete sentence does not repair an earlier paragraph context gap.

The 2013 induction paper is a useful warning about interpretation of numbers. [EPH13, §5, Table 2] reports results on 200 generated short sentences with a 90/10 split, excludes unseen test words, and distinguishes parse coverage from formula accuracy. Its high top-three coverage is not evidence for general unrestricted-text accuracy. Likewise our passing regression tests are not a new held-out accuracy figure.

Do not reduce “verified” to “the root has a type.” Test whether the derivation consumes the intended words, preserves both sides of a relation, binds the intended variables, retains referential distinctions, and exposes contextual assumptions. Completion, semantic typing and empirical adequacy should remain separate fields in the application.

9 Bibliography and review ledger

Page numbers below distinguish physical PDF pages from printed page numbers. “Inspected” indicates targeted review of the relevant passages, not an assertion that every derivation in a long source was independently checked.

[CKM05] Ronnie Cann, Ruth Kempson and Lutz Marten. *The Dynamics of Language: An Introduction*. Elsevier, 2005. Inspected version: 12 December 2004 final draft, 386 PDF pages, local file cann-et-al-dec12 (1).pdf. Contents and selected passages in chs.2–9; especially printed pp.74–80 (PDF84–90), §§3.2–3.3, §5.4, §§6.1–6.2, ch.7, §8.3 and §9.2. Publication-level and draft-level pagination must not be conflated.

[HG21] Christine Howes and Hannah Gibson. “Dynamic Syntax: The Dynamics of Incremental Processing: Constraints on Underspecification.” *Journal of Logic, Language and Information* 30, 263–276,

2021. <https://doi.org/10.1007/s10849-021-09334-x>. Full article inspected. This provides both a contemporary framework overview and explicitly secondary pointers to papers in the special issue.

[C10] Stergios Chatzikyriakidis. *Clitics in Four Dialects of Modern Greek: A Dynamic Account*. PhD thesis, King’s College London, 2010. 396-page local PDF [chatzikyriakidis-phdthesis.pdf](#). Contents, §2.2.2, selected clitic/PCC material and §3.4.3.2 inspected. Detailed future feature work must return to each dialect’s conditions, not only the abstract.

[CK11] Stergios Chatzikyriakidis and Ruth Kempson. “Standard Modern and Pontic Greek Person Restrictions: A Feature-Free Dynamic Account.” *Journal of Greek Linguistics* 11, 127–166, 2011. <https://doi.org/10.1163/156658411X599983>. Inspected locally as [jgl-article-p127_3.pdf](#), particularly the structural/case account at printed pp.144–145. This is evidence for that account, not a new population study of present-day acceptability.

[KC16] Ruth Kempson, Ronnie Cann, Eleni Gregoromichelaki and Stergios Chatzikyriakidis. “Language as Mechanisms for Interaction.” *Theoretical Linguistics* 42(3–4), 203–275, 2016. <https://doi.org/10.1515/tl-2016-0011>. Inspected version: 52-page draft dated 21 April 2016, [tling-draft.pdf](#), selected sections on growth, context, interaction and references. Draft and final pagination differ.

[KE19] Ruth Kempson, Eleni Gregoromichelaki, Arash Eshghi and Julian Hough. “Ellipsis in Dynamic Syntax.” In *The Oxford Handbook of Ellipsis*. <https://doi.org/10.1093/oxfordhb/9780198712398.013.9>. Crossref records publication in 2019; some bibliographies cite 2018. Inspected version: 24-page draft explicitly dated 4 October 2016, locally named [KempsonEtAl18Ellipsis_preprint.pdf](#). Relevant discussion: §§1.2–1.3, especially PDF13–16 on reuse, PDF17–20 on interaction, PDF20–21 on islands. The filename is not treated as the draft’s publication date.

[PEH11] Matthew Purver, Arash Eshghi and Julian Hough. “Incremental Semantic Construction in a Dialogue System.” IWCS 2011, 365–369. <https://aclanthology.org/W11-0144/>. Full five-page paper inspected; §2 describes DS-TTR/event semantics and §3 the DyLan implementation.

[E15] Arash Eshghi, Christine Howes, Eleni Gregoromichelaki, Julian Hough and Matthew Purver. “Feedback in Conversation as Incremental Semantic Update.” IWCS 2015, 261–271. <https://aclanthology.org/W15-0130/>. Inspected §§2–3, especially record dependencies, parsing DAGs, coordination pointers and clarification.

[EPH13] Arash Eshghi, Matthew Purver and Julian Hough. “Probabilistic Induction for an Incremental Semantic Grammar.” IWCS 2013. <https://aclanthology.org/W13-0110/>. Inspected abstract, §§4.4–4.6 and evaluation §5. Learning assumptions and generated-data limitations are retained in the discussion above.

[ECH13] Arash Eshghi and colleagues. “Incremental Grammar Induction from Child-Directed Dialogue Utterances.” Workshop on Cognitive Modeling and Computational Linguistics, 2013, 94–103. <https://aclanthology.org/W13-2611/>. Title/abstract and bibliographic identification inspected from the local paper. Included as an implemented research direction; this report makes no numerical accuracy claim for it.

[C17] Stergios Chatzikyriakidis. “Modification in Dynamic Syntax.” 2017 conference abstract, two content pages plus a repository cover. <https://www.researchgate.net/publication/313634514>. Full abstract inspected. It explicitly motivates alternatives; it is not treated as a full published grammar or a consensus analysis of every modifier class.

[C25] Stergios Chatzikyriakidis. “Constructive Dynamic Syntax.” *Languages* 10(11), 269, 2025. <https://doi.org/10.3390/languages10110269>. Publisher PDF: https://mdpi-res.com/d_attachment/languages/languages-10-00269/article_deploy/languages-10-00269.pdf. Inspected §§3–4 and conclusion, with particular attention to nominal typing, Σ/Π , §4.3 action contexts and modifier alternatives. A foundational proposal with worked examples, not an unrestricted implemented grammar.

[KV19] Ruth Kempson and colleagues. “Why Natural Language Models Must Be Partial and Shifting: A Dynamic Syntax with Vector Space Semantics Perspective.” Local two-page IWCS 2019 contribution

kempson.etal_IWCS_19.pdf. Title and proposal scope inspected. This motivates an architectural option; no distributional backend is added to this app.

9.1 Sources to acquire or review directly next

- Kempson, Meyer-Viol and Gabbay (2001), *Dynamic Syntax: The Flow of Language Understanding*: the foundational monograph, repeatedly cited in the reviewed sources. This review uses [CKM05] and [HG21] for directly inspected mechanisms.
- Ronnie Cann (2011), “Towards an Account of the English Auxiliary System,” in *The Dynamics of Lexical Interfaces*, pp.279–317: identified in [KC16].
- Lutz Marten (2002), *At the Syntax-Pragmatics Interface: Verbal Underspecification and Concept Formation in Dynamic Syntax*: identified in [C17], relevant to valency/modification.
- Eleni Gregoromichelaki (2006), *Conditionals: A Dynamic Syntax Account*: identified in [HG21] and [KE19]; necessary before claiming the workbench implements the theoretical conditional analysis.
- Miriam Bouzouita (2008), *The Diachronic Development of Spanish Clitic Placement*: identified in [KC16].
- Kempson and Kiaer (2010), “Multiple Long-Distance Scrambling: Syntax as Reflections of Processing,” *Journal of Linguistics* 46, 127–192; Gibson (2016), “A Unified Dynamic Account of Auxiliary Placement in Rangi,” *Lingua* 184, 79–103; Seraku and Gibson (2016), “A Dynamic Syntax Modelling of Japanese and Rangi Clefts,” *Language Sciences* 56, 45–67: identified in [HG21].
- Japanese case, Kazakh differential object marking and Mandarin minimizers papers summarized in [HG21, §3]: secondary evidence only at this stage.
- “Dynamic Syntax,” *The Oxford Handbook of Philosophy of Linguistics*, 2026, <https://doi.org/10.1093/oso/9780198879640.003.0020>: metadata verified; chapter contents not reviewed.

10 Relation to the project foundation

The software foundation is the `dynamic-syntax` Python package and its `dylan` engine, derived from the Dylan implementation tradition documented in [PEH11]. This project builds on its action interpreter, trees, records and context/search machinery. The workbench adds native Classical/Constructive compilation, grammar families, diagnostics, operation playback, paragraph and dialogue flows, source investigation, bounded model assistance and visible Jev comparisons. The new relative family is another explicit grammar/semantic extension on that foundation.

The research programme does not relabel those additions as inherited features. Equally, the original literature’s implemented capabilities are not automatically advertised as working in this port: they need regression evidence in the selected grammar and semantic backend.